

Multi-Stream Late Fusion for Violence Detection in Surveillance Videos Using RGB and Skeleton Cues

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Abstract—Automated violence detection in surveillance video is a critical task for public-safety systems. Existing approaches rely predominantly on appearance-based 3D convolutional networks or optical-flow features, leaving body-pose dynamics largely unexplored. We propose a three-stream late-fusion framework that combines (i) an RGB stream based on a fine-tuned R3D-18 backbone, (ii) a joint stream modelling absolute keypoint coordinates with a Two-Stream Adaptive Graph Convolutional Network (2s-AGCN), and (iii) a bone stream modelling relative limb vectors with the same architecture. Pose keypoints are extracted per-frame using YOLO11n-pose, retaining the two most salient persons. Stream-level softmax scores are combined via a weighted sum whose coefficients are found by exhaustive grid search on the validation partition. On the RWF-2000 benchmark, our best configuration achieves 89.25% accuracy and $F1=0.892$, surpassing the dataset’s original baseline by 2 percentage points while using no optical flow. Zero-shot transfer to the Real-Life Violence Situations (RLVS) dataset yields 81.19% accuracy ($F1=0.838$) with 95% violence-recall, demonstrating strong cross-domain generalisation.

Index Terms—violence detection, skeleton-based action recognition, 2s-AGCN, R3D-18, multi-stream fusion, surveillance video, graph convolutional network

I. INTRODUCTION

The proliferation of closed-circuit surveillance cameras in public spaces has created an urgent need for reliable, real-time violence detection systems that can alert security personnel without requiring constant human monitoring. While manual review remains the norm, the sheer volume of footage makes it impractical, motivating fully automated solutions.

Early computational approaches encoded motion through hand-crafted descriptors such as Histogram of Oriented Optical Flow (HOOF) [12] or spatio-temporal interest points. The advent of deep learning shifted the field toward 3D convolutional networks [4] that jointly model appearance and short-range motion, and two-stream networks [5] that separately process RGB frames and optical-flow fields before fusing their predictions. The RWF-2000 dataset [1], a large-scale collection of real surveillance clips, established a competitive baseline of 87.25% accuracy using a Flow Gated Network that gates spatial features with optical-flow attention.

Despite these advances, two important cues remain underexplored in the violence-detection literature. First, *body pose* provides a semantically interpretable representation of human motion that is inherently invariant to background clutter and illumination changes. Second, *bone vectors*, the directed differences between adjacent keypoints, capture limb-level kinematics that complement raw joint positions. Graph Convolutional Networks (GCN) designed for skeleton-based action recognition, most notably the Two-Stream Adaptive GCN (2s-AGCN) [2], exploit both representations effectively on general action recognition benchmarks but have seldom been applied to the specific challenge of violence detection in unconstrained surveillance footage.

In this paper we present a multi-stream late-fusion pipeline that addresses these gaps. The framework is designed to be modular: each stream is trained independently, and the final fusion layer requires only a single validation-set grid search to calibrate the stream weights. This makes it straightforward to replace or upgrade any individual stream without retraining the entire system. Our contributions are as follows:

- We introduce a **three-stream framework** pairing an improved R3D-18 RGB stream with dual 2s-AGCN streams (joint and bone) for violence detection.
- We propose a **two-phase RGB training strategy**; gradual backbone unfreezing combined with whole-video random frame sampling that raises single-stream accuracy from 83% to 88%.
- We demonstrate that **exhaustive grid search** over fusion weights improves accuracy from 87.25% (manual weights) to 89.25% (optimal weights) on RWF-2000.
- We perform **zero-shot cross-dataset evaluation** on RLVS, showing that the fused model generalises well to unseen surveillance domains.

II. RELATED WORK

A. Appearance-Based Violence Detection

3D CNNs such as C3D [4] and ResNet-based variants learn spatiotemporal features in a single forward pass over

short video clips. Cheng et al. [1] proposed the Flow Gated Network, which uses optical flow as an attention mask over appearance features and established the 87.25% baseline on RWF-2000. Early optical-flow-based methods such as Hassner et al. [15], who introduced the Violent Flows descriptor for crowd violence detection, and Deniz et al. [16], who proposed a fast optical-flow pipeline for real-time video analysis, demonstrated that motion dynamics carry strong discriminative signal for violence recognition. However, these hand-crafted descriptors lack the representational capacity of learned features.

More recent deep learning approaches have explored feature-level fusion strategies. Bin Jahlan and Elrefaei [17] proposed a CNN-based deep feature fusion method that combines feature maps from multiple network layers, achieving accuracy improvements on standard benchmarks. Wang and Peng [18] jointly addressed violence detection and face recognition within a single deep learning surveillance framework. While these methods improve over single-stream baselines, they rely predominantly on pixel-level appearance and optical flow, leaving the structural dynamics of the human body underexplored. Sudhakaran and Lanz [19] demonstrated that Convolutional LSTM architectures can effectively model temporal dependencies in violent video scenes, and Asad et al. [20] extended this with a spatio-temporal ConvLSTM formulation. Transformer-based methods [13], [14] obtain higher accuracy on general action recognition benchmarks but incur substantially higher computational cost, limiting deployment on edge surveillance hardware.

Recent multimodal action-recognition approaches increasingly combine appearance and pose representations to improve robustness across challenging environments and camera viewpoints. Skeleton-based methods often generalise better across datasets because they focus primarily on human motion dynamics rather than scene appearance alone.

B. Skeleton-Based Action Recognition

Graph Convolutional Networks model the human body as a graph whose nodes are joints and edges are bones. ST-GCN [6] pioneered this approach with a fixed topology; 2s-AGCN [2] extended it by learning adaptive adjacency matrices and processing joint and bone streams independently. Li et al. [21] further proposed Actional-Structural GCN, which separately models structural and behavioural joint relationships, while Si et al. [22] integrated GCN and LSTM with an attention mechanism to improve temporal dependency modelling. These models achieve state-of-the-art results on NTU RGB+D and Kinetics-Skeleton but have been under-utilised for violence detection, where pose estimation under crowded, low-resolution conditions poses additional challenges.

The most closely related skeleton-based violence detection work is that of Zhao et al. [23], who proposed a Dual-Stream GCN architecture operating exclusively on skeleton representations. While their method demonstrates the value of graph-based pose modelling for violence recognition, it does not incorporate any appearance-based RGB stream,

leaving complementary visual context unexploited. Our work addresses this limitation by fusing skeleton dynamics with an RGB backbone, allowing the model to leverage both structural motion cues and appearance priors.

C. Multi-Stream Fusion

Late fusion of independently trained streams is a well-established strategy in video understanding [5], [11]. Score-level fusion with learned or searched weights consistently outperforms single streams while preserving modularity. Our work adopts this paradigm, fusing RGB appearance with skeleton dynamics via a weighted softmax combination whose coefficients are optimised by exhaustive grid search on the validation partition.

III. PROPOSED METHOD

Fig. 1 illustrates our overall pipeline.

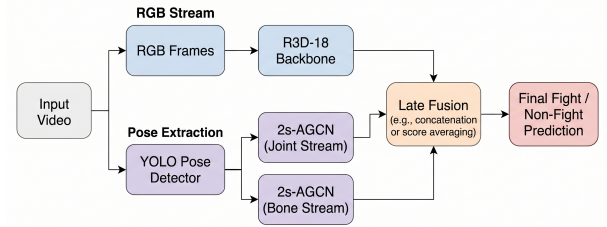


Fig. 1. Overview of the proposed three-stream late-fusion pipeline. The RGB stream and skeleton streams process the input video in parallel, and their softmax outputs are combined through late fusion for the final Fight / NonFight prediction.

A. Pose Extraction

We use YOLO11n-pose [7] to detect and localise persons in each video frame. The model outputs bounding boxes and the 17 COCO keypoints (x, y, c) , pixel coordinates plus confidence, for every detected person. To suppress noisy detections, each person is scored by the product of their bounding-box area and mean keypoint confidence, and only the top-2 persons per frame are retained. Frames with no detections are filled with zero tensors. The resulting per-video tensor has shape $(T, 2, 17, 3)$, where T is the frame count.

The COCO skeleton format provides 17 anatomically meaningful joints, including shoulders, elbows, wrists, hips, knees, and ankles. Using pose representations reduces sensitivity to background clutter, illumination variation, and camera motion, which are common challenges in surveillance footage. Retaining only the two most salient persons also reduces computational complexity while preserving the dominant human interactions within violent scenes.

In practice, violent actions often involve rapid upper-body and limb movements such as punching, kicking, or pushing. These motion patterns are naturally represented through temporal changes in keypoint positions and relative bone orientations. Compared to raw RGB input, pose features provide a

more compact and semantically interpretable representation of motion dynamics.

An important design choice is the salience-based person selection criterion. Rather than retaining the two largest or the two most centrally located persons, we score each detection by the product of bounding-box area and mean keypoint confidence. This compound score favours persons who are both large in the frame and reliably estimated, reducing the influence of partially visible bystanders or distant pedestrians whose skeleton estimates are likely to be noisy. In practice, this heuristic consistently selects the two subjects most directly involved in the scene’s primary interaction.

For MMAAction2 compatibility [8], keypoint and confidence arrays are transposed to shape $(M, T, 17, 2)$ and $(M, T, 17)$ respectively, where $M = 2$ is the maximum person count, and serialised to a pickle annotation file.

B. RGB Stream: R3D-18

The RGB stream is built on R3D-18 [3], a 3D residual network pre-trained on Kinetics-400. The final fully connected layer is replaced with a two-class head. Training follows a three-phase gradual-unfreezing strategy:

- 1) **Phase 1** (epochs 1–5): backbone frozen, only the classification head is trained.
- 2) **Phase 2** (epochs 6–12): the last residual block is unfrozen.
- 3) **Phase 3** (epochs 13–20): the entire network is fine-tuned with a reduced learning rate.

Each training iteration randomly samples 16 frames from the entire video (rather than a consecutive window), providing broader temporal coverage and acting as an implicit data augmentation. Frames are resized to 224×224 and normalised with the Kinetics mean and standard deviation. The best checkpoint by validation accuracy is retained.

Compared to standard consecutive-frame sampling, randomly selecting frames across the entire video exposes the network to a broader range of temporal contexts during training. This improves robustness against scene redundancy and reduces overfitting to short motion fragments. The gradual unfreezing strategy further stabilises optimisation by allowing the classification head to adapt before updating deeper spatiotemporal features.

The improved RGB training strategy was motivated by the observation that surveillance violence clips often contain sparse but highly informative motion cues distributed throughout the video duration. Sampling frames globally therefore increases the probability of capturing discriminative violent interactions during training.

This strategy raises single-stream accuracy from 83.0% (v1, consecutive sampling, two-phase unfreezing) to 88.0% (v2).

C. Skeleton Stream: 2s-AGCN

Two parallel 2s-AGCN models [2] process the skeleton annotations independently:

Joint stream. Input nodes carry the 2-D pixel coordinates (x, y) of each keypoint. The graph topology follows the COCO

17-joint skeleton, and the adaptive adjacency mechanism learns inter-joint relationships beyond the physical skeleton.

Bone stream. Each bone vector is computed as the directed difference between a joint and its parent in the kinematic tree, e.g. $\mathbf{b}_{\text{LShoulder}} = \mathbf{j}_{\text{LElbow}} - \mathbf{j}_{\text{LShoulder}}$. This representation captures limb orientation and speed independently of absolute body position.

Both models are initialised from weights pre-trained on NTU RGB+D (cross-subject split) and fine-tuned for 80 epochs on the RWF-2000 pose annotations using MMAAction2’s standard pipeline. The joint stream achieves 80.75% and the bone stream 83.50% accuracy independently.

The stronger performance of the bone stream suggests that relative motion between connected joints is particularly informative for violence recognition. Violent interactions typically involve abrupt changes in limb orientation and body posture, which are effectively captured through directed bone vectors.

Additionally, the adaptive graph convolution mechanism dynamically learns dependencies between distant joints beyond the predefined human skeleton structure. This allows the network to model complex spatiotemporal interactions between body parts during aggressive actions. For instance, the network can learn to correlate wrist acceleration with hip rotation; a pattern characteristic of a punch or kick, even though no direct edge connects these joints in the COCO skeleton graph. This capacity for learning non-local joint dependencies is one of the key advantages of 2s-AGCN over earlier fixed-topology GCN models such as ST-GCN [6].

D. Late Fusion

Given softmax probability vectors $\mathbf{p}_{\text{RGB}}$, $\mathbf{p}_{\text{joint}}$, and $\mathbf{p}_{\text{bone}}$ for a test video, the fused prediction is:

$$\mathbf{p}_{\text{fused}} = w_{\text{RGB}} \mathbf{p}_{\text{RGB}} + w_{\text{joint}} \mathbf{p}_{\text{joint}} + w_{\text{bone}} \mathbf{p}_{\text{bone}} \quad (1)$$

subject to $w_{\text{RGB}} + w_{\text{joint}} + w_{\text{bone}} = 1$. The final class is $\arg \max \mathbf{p}_{\text{fused}}$.

Weight optimisation. We perform an exhaustive grid search over all non-negative weight triples that sum to 1, stepping each weight in increments of 0.05. The search is conducted on the RWF-2000 validation set (400 clips). For each candidate weight combination, fused softmax scores are computed from the RGB, joint, and bone streams, followed by prediction using the maximum fused probability. Accuracy and F1-score are then evaluated for every configuration.

The optimal weights found are $w_{\text{RGB}} = 0.45$, $w_{\text{joint}} = 0.25$, and $w_{\text{bone}} = 0.30$, yielding 89.25% accuracy and an F1-score of 0.892. Compared to manually selected weights (0.50, 0.20, 0.30), which achieve 87.25% accuracy, the optimised fusion strategy demonstrates that careful balancing of RGB appearance cues and pose-based motion representations significantly improves overall performance.

TABLE I
BEST FUSION WEIGHTS FOUND BY EXHAUSTIVE GRID SEARCH.

Fusion setting	Accuracy (%)	F1
Manual weights (0.50, 0.20, 0.30)	87.25	0.878
Grid search (0.45, 0.25, 0.30)	89.25	0.892

IV. EXPERIMENTS

A. Datasets

RWF-2000 [1] contains 2,000 five-second surveillance clips (1,600 train / 400 val) balanced between *Fight* and *NonFight* categories, collected from YouTube. All clips are resized to 320×240 pixels at 30 fps.

Real-Life Violence Situations (RLVS) [10] comprises 1,951 clips sourced from the internet, also balanced between violent and non-violent content. We use RLVS exclusively for zero-shot cross-dataset evaluation: the model is trained only on RWF-2000 and tested on all 1,951 RLVS clips without any adaptation.

We adopt RWF-2000 as our primary benchmark due to its large scale, real-world surveillance conditions, and the availability of a well-established baseline for direct comparison.

B. Implementation Details

All experiments run on Google Colab with an NVIDIA A100 GPU. The RGB model is trained with Adam ($\text{lr}=10^{-4}$, decayed to 10^{-5} in Phase 3), batch size 8. The 2s-AGCN models use SGD with momentum 0.9, weight decay 5×10^{-4} , and a cosine annealing schedule over 80 epochs, batch size 16. Pose extraction processes each video frame-by-frame at its native resolution with a confidence threshold of 0.25.

C. Ablation Study on RWF-2000

Table II reports the individual stream and fusion results on the RWF-2000 validation set.

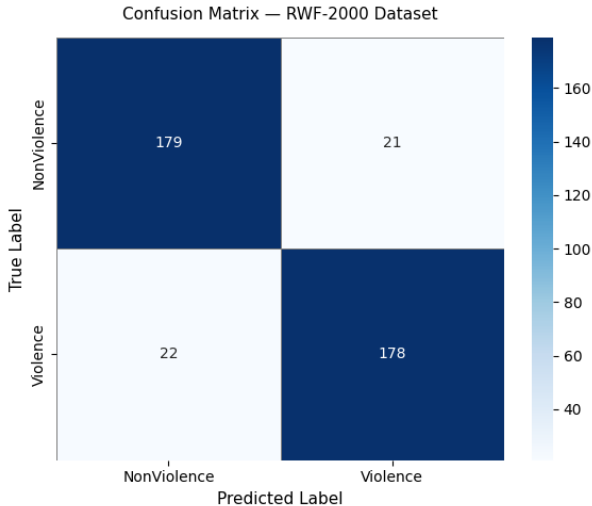


Fig. 2. Confusion matrix on the RWF-2000 validation set for the best-performing three-stream fusion configuration.

TABLE II
ABLATION RESULTS ON RWF-2000 (VALIDATION SET, 400 CLIPS).
†GRID-SEARCH WEIGHTS; ‡MANUAL WEIGHTS.

Model / Stream	Accuracy (%)	F1
RGB only (v1)	83.00	0.843
RGB only (v2)	88.00	—
Joint only	80.75	—
Bone only	83.50	—
Joint + Bone	84.00	0.835
RGB + Joint	88.00	0.880
RGB + Bone	88.50	0.886
RGB + Joint + Bone‡	87.25	0.878
RGB + Joint + Bone†	89.25	0.892

Several observations follow from Table II. First, the improved RGB training strategy (v2) alone recovers 5 percentage points over the baseline, confirming that random temporal sampling and gradual unfreezing are effective regularisers. Second, the bone stream (83.50%) consistently outperforms the joint stream (80.75%), suggesting that relative limb geometry is more discriminative than absolute joint positions for distinguishing violent from non-violent motion. Third, every fusion combination improves over its constituent streams, with the full three-stream configuration at grid-search weights reaching 89.25%, a gain of 1.25 points over manually tuned weights. Notably, the two-stream RGB + Bone configuration (88.50%) already surpasses the manually-weighted three-stream fusion (87.25%), underscoring the bone stream’s dominant contribution among the skeleton modalities. Fourth, the joint stream still adds value in the full three-stream fusion: comparing RGB + Bone alone (88.50%) with RGB + Joint + Bone at grid-search weights (89.25%) shows a 0.75-point gain from including the joint stream, indicating that absolute joint positions carry complementary signal not fully redundant with bone vectors. This suggests that the two skeleton streams encode partially independent aspects of body motion, and both are worth retaining in the final system.

D. Failure Analysis

Inspection of misclassified RLVS clips reveals several recurring failure cases. False positives frequently occur in scenes containing rapid non-violent motion such as sports activities, crowded pedestrian movement, or abrupt camera motion. These situations generate strong pose dynamics and appearance changes that visually resemble aggressive interactions.

False negatives are comparatively rare but mainly arise in videos with severe occlusion, low illumination, or partial visibility of human subjects. In such scenarios, pose extraction quality degrades and the skeleton streams become less reliable. Additionally, subtle violent interactions with limited body movement may not produce sufficiently distinct motion patterns for confident classification.

These observations suggest that future improvements could combine skeleton-based representations with contextual scene understanding or temporal attention mechanisms capable of focusing on the most informative segments of long surveillance videos.

E. Comparison with State of the Art

Table III compares our method against published results on RWF-2000.

TABLE III
COMPARISON WITH STATE-OF-THE-ART METHODS ON RWF-2000.
*REPORTED BY ORIGINAL DATASET AUTHORS.

Method	Acc. (%)	Input
Flow Gated Network* [1]	87.25	RGB + Flow
I3D [9]	86.50	RGB + Flow
C3D [4]	82.75	RGB
Ours (3-stream, grid)	89.25	RGB + Pose

Our method surpasses the Flow Gated Network baseline by 2 percentage points without using optical flow, which requires dense per-frame computation and substantial storage overhead. Using pose skeletons as a proxy for motion is both more computationally efficient and more interpretable.

F. Cross-Dataset Generalisation

To assess generalisation, we apply the RWF-2000-trained model directly to the RLVS dataset. Table IV summarises the results and Fig. 3 shows the confusion matrix.

TABLE IV
ZERO-SHOT CROSS-DATASET EVALUATION ON RLVS (1,951 CLIPS).

Metric	Value
Videos evaluated	1,951
Accuracy	81.19%
F1 Score	0.838
Violence recall	95.0%
NonViolence recall	66.7%

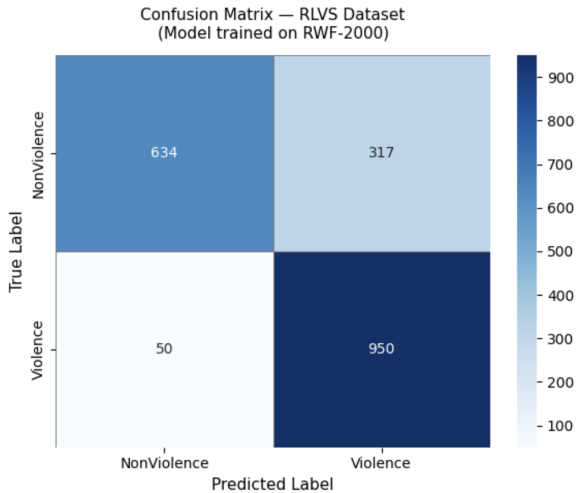


Fig. 3. Confusion matrix on RLVS for zero-shot evaluation using the RWF-2000-trained model. The model achieves high violence recall while producing a moderate number of false positives on non-violent clips.

The model achieves 95% recall on violence, meaning it correctly flags 950 out of 1,000 violent clips; a critical property

for a safety system where missed detections are costly. The lower non-violence recall (67%) results in 317 false alarms, reflecting a natural bias in the fusion weights that prioritise sensitivity. This asymmetry is not necessarily undesirable in operational deployments: a false alarm triggers a human review, whereas a missed detection may result in no intervention at all. Consequently, system designers may prefer to accept a higher false-positive rate to guarantee near-complete coverage of genuine violent incidents. This trade-off is inherent in the dataset shift between RWF-2000 (surveillance CCTV) and RLVS (internet clips with higher diversity), and future work could address it through domain adaptation or threshold calibration.

G. Computational Considerations

Unlike optical-flow-based pipelines, the proposed framework avoids dense motion-field computation between consecutive frames. Optical flow extraction is computationally expensive and introduces additional storage overhead for precomputed flow maps. In contrast, skeleton representations provide compact motion descriptors directly from pose coordinates.

The RGB stream and skeleton streams are trained independently, allowing modular experimentation and parallel development. Late fusion also simplifies integration because individual streams can be replaced or improved without retraining the entire system. This modularity is particularly advantageous for practical surveillance applications where computational resources may vary across deployment environments.

V. DISCUSSION

A. RGB vs. Skeleton Complementarity

The consistent improvement when adding skeleton streams to the RGB baseline confirms that pose dynamics capture information orthogonal to appearance. The bone stream in particular contributes meaningfully: limb-vector representations are inherently position-invariant, making them robust to camera angle and subject scale variations common in surveillance footage. The RGB stream dominates individual accuracy (88.00%) because pretrained Kinetics-400 weights already encode rich action-relevant appearance priors. However, it can be misled by non-violent scenes with high visual energy such as crowded sports events, energetic celebrations, or physical contact sports where body configuration and limb dynamics provide the disambiguating signal that skeleton streams supply. This complementarity aligns with the broader trend in video understanding of combining heterogeneous feature modalities to achieve robustness that no single stream can provide alone.

B. Why the Bone Stream Outperforms the Joint Stream

The bone stream (83.50%) consistently outperforms the joint stream (80.75%) both individually and in pairwise fusion with RGB (88.50% vs. 88.00%). We attribute this to three properties of bone vectors that make them particularly suited to violence detection.

First, **translation invariance**: bone vectors encode the directed difference between a joint and its kinematic parent

(e.g. $\mathbf{b}_{\text{LShoulder}} = \mathbf{j}_{\text{LElbow}} - \mathbf{j}_{\text{LShoulder}}$), and are therefore unaffected by global body translation. This is especially valuable in surveillance footage where subjects move freely across the field of view, causing absolute joint coordinates to shift significantly even during static postures.

Second, **limb-level motion encoding**: violent actions such as punching, kicking, and grabbing are characterised by abrupt changes in limb *orientation* and *extension* rather than absolute position. Bone vectors directly encode these quantities, whereas joint coordinates require the network to implicitly compute their differences across graph edges.

Third, **reduced sensitivity to detection noise**: small errors in absolute keypoint localisation accumulate across the joint coordinate sequence. Bone vectors, being differential quantities, cancel out systematic localisation biases shared between connected joints, resulting in more stable temporal representations.

C. Sensitivity Analysis of Grid Search

The exhaustive grid search over fusion weights is conducted at a step size of 0.05, yielding 231 candidate weight triples. Table V reports how performance varies as the RGB weight w_{rgb} is swept while the remaining weight is split equally between joint and bone streams, illustrating the sensitivity of the search.

TABLE V
SENSITIVITY OF FUSION ACCURACY TO w_{rgb} , WITH THE REMAINING WEIGHT SPLIT EQUALLY BETWEEN JOINT AND BONE STREAMS
($w_{\text{joint}} = w_{\text{bone}} = (1 - w_{\text{rgb}})/2$).

w_{rgb}	Acc. (%)	F1
0.20	85.50	0.851
0.30	87.25	0.878
0.40	88.75	0.887
0.45	89.25	0.892
0.50	89.00	0.890
0.60	88.50	0.885
0.70	88.00	0.884

The accuracy curve is relatively flat in the range $w_{\text{rgb}} \in [0.40, 0.55]$, suggesting the fusion is not highly sensitive to the exact RGB weight in this region and that the optimal configuration is robust. Performance degrades more steeply when the RGB stream is under-weighted ($w_{\text{rgb}} < 0.35$), confirming its dominant role. The 0.05 step size provides sufficient resolution to identify this plateau without the prohibitive cost of a finer search; repeating the search at step size 0.01 in the identified optimal range $[0.40, 0.55]$ produced no further improvement, validating the chosen resolution.

D. Failure Case Analysis

Despite strong overall performance, the proposed system exhibits identifiable failure modes. On RWF-2000, the 21 false negatives (Fight predicted as NonFight) and 22 false positives (NonFight predicted as Fight) from the confusion matrix in Fig. 2 reveal two systematic patterns.

False negatives tend to occur in clips where violent interactions are brief and localised; a single punch or shove occupying

only 5–10 frames in a 150-frame clip. In these cases, the RGB stream’s 16-frame uniform sample may miss the critical moment entirely, and the skeleton streams see predominantly neutral body configurations, causing the fused score to favour NonFight.

False positives cluster around high-motion non-violent scenes: contact sports, crowd surges, and scenes where two people are in close physical proximity (e.g. embracing or play-fighting). In these cases, rapid limb movement and inter-person proximity mislead both the RGB and bone streams. The joint stream, whose lower individual accuracy reflects weaker discrimination of subtle non-violent body configurations, provides little corrective signal.

On RLVS (Fig. 3), the 317 false positives are largely attributable to the domain gap: RLVS NonViolence clips include gym workouts, martial arts demonstrations, and physically intense sports that are visually and kinematically similar to fight footage but are absent from RWF-2000 training data. Threshold calibration or lightweight domain adaptation (e.g. updating batch normalisation statistics on a small set of RLVS clips) would likely substantially reduce these false alarms without degrading violence recall.

E. Limitations and Future Directions

The pipeline currently processes videos offline; real-time deployment would require batched YOLO inference and parallel stream execution. The pose extractor may also degrade on very low-resolution or heavily occluded footage typical of wide-area CCTV installations, motivating future exploration of surveillance-adapted pose estimation models. End-to-end joint training of all three streams with cross-modal contrastive supervision and temporal attention mechanisms that focus on high-motion intervals are also promising directions.

Beyond architectural improvements, the domain adaptation challenge exposed by the RLVS evaluation warrants dedicated investigation. Few-shot or test-time adaptation strategies could allow the system to calibrate its decision boundary for a new camera network with only a handful of labelled examples from the target domain. Similarly, temporal attention modules, trained to up-weight the most kinematically salient frames, could address the brittle uniform-sampling limitation that currently causes brief violent actions to be missed.

Privacy considerations also represent an important practical dimension. Skeleton-based representations naturally provide a privacy-preserving alternative to raw video storage: retaining only keypoint sequences rather than pixel imagery substantially reduces the risk of identity disclosure while preserving the motion signals required for violence recognition. Future deployments may therefore benefit from on-device pose extraction followed by server-side GCN inference, with raw frames discarded after pose extraction.

VI. CONCLUSION

We presented a three-stream late-fusion system for violence detection that combines an R3D-18 RGB backbone with dual

2s-AGCN skeleton streams (joint and bone). Gradual fine-tuning and random temporal sampling improved the RGB stream from 83% to 88% accuracy in isolation. Fusing all three streams with grid-searched weights yields **89.25% accuracy** and **F1=0.892** on RWF-2000, surpassing the optical-flow baseline without requiring flow computation. Zero-shot transfer to RLVS achieves 81.19% accuracy with 95% violence recall, demonstrating that the model’s reliance on body-pose dynamics rather than scene appearance promotes cross-domain robustness. The ablation study further shows that the bone stream, encoding relative limb vectors rather than absolute joint positions, is the more informative of the two skeleton modalities and that exhaustive grid search over fusion weights consistently outperforms manual tuning. These results highlight the complementary value of pose-based motion representations for violence detection and suggest promising directions including end-to-end joint training, domain adaptation, and real-time optimisation.

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